Improving Lightweight Super-Resolution via Depth Compression: A Statistically-Validated Accuracy–Efficiency Trade-off for Ear Biometrics

Thai Le Quang, Vinh Truong Hoang*

Faculty of Information Technology, Ho Chi Minh City Open University

thailq.25ai@ou.edu.vn, vinh.th@ou.edu.vn

Abstract

Ear recognition is gaining attention as a complement to face recognition when the face is occluded or unavailable. In practice, ear images are often captured at very low resolution, for example in surveillance footage, distant capture, or on resource-constrained edge cameras. Lightweight super-resolution (SR) networks such as SPAN (Swift Parameter-free Attention Network) offer a way to pre-process such images, but they were designed and validated at resolutions well above the 20×20-pixel regime typical of distant ear capture, and it remains unclear how much further they can be compressed before downstream recognition accuracy is affected. This paper proposes **span_tiny**, a depth-compressed variant of SPAN with 3 attention blocks instead of 6, giving 0.230M parameters and a 46.0% reduction from the uncompressed baseline. Five recognition backbones are evaluated with $n = 5$ seeds and paired statistical testing (paired t-test, Cohen's d, Bonferroni correction). On the primary dataset (EarVN1.0, 164 subjects), both span_tiny and the uncompressed baseline significantly beat the no-SR condition in all five backbones (Cohen's d 2.2–10.6), while the accuracy cost of compression itself reaches significance in only one of five backbones (EfficientNet-B0, 1.2 percentage points) and stays under one percentage point in the remaining four. A depth ablation isolating block count alone corroborates this: span_tiny is statistically indistinguishable from an otherwise identical, uncompressed-depth counterpart in four of five backbones. On a second, independent dataset (AWEx, 336 subjects), the same compression-safety finding replicates across three independent evaluation protocols – from-scratch training, full-fine-tuning transfer from EarVN1.0, and frozen-backbone transfer – with zero of five backbones showing a significant accuracy cost from compression under any protocol. Against five other lightweight SR architectures trained under an identical distillation recipe, span_tiny is the second-fastest and third-smallest, is statistically tied with the two most accurate alternatives, and significantly outperforms the two least accurate ones on downstream accuracy, despite ranking low on raw PSNR/SSIM – evidence that pixel-fidelity metrics are a poor proxy for recognition-relevant quality at this compression level. Three further mechanisms proposed to improve on span_tiny – a multi-judge identity/feature-distillation loss, a saliency-weighted identity-critical loss, and learned/differentiable block pruning – are each validated to the same $n = 5$-seed, five-backbone standard and found to provide no reliable improvement, with learned pruning performing significantly worse than span_tiny's fixed block selection in all five backbones. A further ablation testing whether SPAN's parameter-free attention specifically explains its depth tolerance finds no support for that account: two architectures with learned attention mechanisms tolerate an equivalent depth reduction just as well. Results that do not reach statistical significance are reported as such throughout, and an auxiliary check on genuinely low-resolution real-world images (as opposed to synthetically downsampled ones) surfaces a boundary condition on the main claim, traced to a resolution mismatch between these images and the synthetic degradation used to train the SR models.

*Corresponding author (Vinh Truong Hoang): vinh.th@ou.edu.vn

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1 Introduction

1.1 Background

Ear biometrics has drawn growing interest as a complement or alternative to face recognition, particularly since widespread mask-wearing made face-based identification unreliable in many settings [12]. The ear offers several properties useful for identification: its structure remains largely stable across adulthood, it carries a distinctive pattern of cartilage folds, and it changes little with facial expression, unlike the face. However, ear recognition in unconstrained settings – surveillance footage, images captured at a distance, footage from resource-constrained edge cameras – routinely produces ear regions at very low pixel resolution, often well below what recognition backbones or SR networks were designed for.

Lightweight SR has advanced considerably in recent years, driven largely by the NTIRE efficient super-resolution challenge series. SPAN [1] is among the most efficient recent designs. It won both the overall-performance and runtime tracks of NTIRE 2024 by replacing learned attention with a symmetric, parameter-free activation applied directly to convolutional outputs. Other designs pursue efficiency through different mechanisms: RLFN [2] simplifies feature aggregation, ECBSR [3] uses hardware-friendly re-parameterized convolutions, SAFMN [4] relies on spatially-adaptive modulation instead of self-attention, and SMFANet [5] combines self-modulation with a lightweight feed-forward design. All of these architectures are designed and benchmarked at input resolutions of roughly 32–64 px or higher, well above the $\sim$20×20-pixel regime addressed in this paper, and none has been evaluated for further depth compression under statistically rigorous validation against a concrete downstream task.

1.2 Research gap

Two gaps motivate this work. First, most lightweight SR papers report PSNR and SSIM on standard benchmarks such as Set5, Set14, Urban100, and DIV2K. These metrics say little about whether a compressed model remains useful as a pre-processing step for a specific recognition task at extremely low input resolution. Some precedent exists for optimizing SR toward a downstream task rather than pixel fidelity: Ataer-Cansizoglu et al. [6] demonstrated this for low-resolution face verification. Almost no equivalent work exists for ear biometrics, and none addresses how far a lightweight architecture can be compressed before downstream accuracy degrades. Second, papers in this area typically report a single training run without a variance estimate, making it difficult to determine whether a reported accuracy gap reflects a genuine effect or seed-to-seed noise. This concern is more acute for downstream classification accuracy than for SR pixel metrics, since classification accuracy is a noisier signal, and a single run therefore provides weaker support for a claim than it would for PSNR.

1.3 Contributions

This paper's central contribution is **span_tiny**, a depth-compressed variant of SPAN (`n_blocks` reduced from 6 to 3, `feat`=48) that cuts parameter count by 46.0% (0.230M vs. 0.4263M) and lowers inference latency correspondingly; the re-implementation is validated against the original authors' published SPAN-S results before being applied to the new task, matching 99.93% of the reported benchmark PSNR. Three further results carry most of the paper's evidentiary weight:

(a) For a single, pre-registered question – does depth compression preserve the ordering `no-SR` < `span_tiny` < `uncompressed baseline`? – multi-seed ($n = 5$), multi-backbone (5 recognition architectures) statistical testing shows that both span_tiny and the uncompressed baseline beat no-SR significantly in all five backbones on the primary dataset, EarVN1.0, while the accuracy cost of compression itself reaches significance in only one.

(b) The same compression-safety finding holds on a second, independent dataset (AWEx, 336 subjects) across three separate evaluation protocols – from-scratch training, full-fine-tuning transfer from EarVN1.0, and frozen-backbone transfer – with no backbone showing a significant accuracy cost under any of the three, evidence of generalization beyond a single dataset or transfer strategy. The secondary question of whether SR helps at all over no-SR is, by contrast, less consistently powered on this smaller dataset, and this is reported directly rather than smoothed over.

(c) A depth ablation that isolates block count alone – span_tiny against span_large, an otherwise identical, uncompressed-depth counterpart – shows no statistically significant difference in four of five backbones on EarVN1.0 (the fifth, EfficientNet-B0, shows a small but significant cost under one percentage point) and in all five on AWEx. This is the strongest single piece of evidence that the depth reduction is close to accuracy-neutral for the task.

Several secondary results round out the picture without carrying the same weight. An ablation of span_tiny's loss components (pixel loss, distillation, and an additional identity term, in all four pairwise combinations) finds that the identity term significantly *reduces* downstream accuracy – consistent with an independent hyperparameter sweep reported in Section 3.3 – while the distillation term's own contribution remains statistically indistinguishable from pixel-only training; both are reported as such, with neither reframed as a positive finding. Set against five other lightweight SR architectures (RLFN, RLFN-adapted, ECBSR, SAFMN, SMFANet) trained under an identical distillation recipe, span_tiny is never significantly beaten on downstream accuracy by any of them: it is statistically tied with the two most accurate alternatives and significantly ahead of the two least accurate, despite ranking low among the six on raw PSNR/SSIM – a sign that this work is competitive on the metric the task actually cares about, not only on efficiency. Three further mechanisms designed to improve on span_tiny beyond its core block-count reduction – a feature-level distillation loss, a saliency-weighted identity-critical loss, and learned/differentiable block pruning – were each validated to the same statistical standard as the main result and found to help not at all: the first two show no detectable effect in either direction, and learned pruning performs significantly *worse* than span_tiny's fixed block selection in every backbone at an equal parameter count, together forming a rigorous negative-result ablation rather than a set of partial successes. Finally, an ablation designed to test this paper's own candidate explanation for why depth reduction is tolerable – that SPAN's attention mechanism carries no learned parameters – finds no support for it: two architectures with learned attention or modulation, subjected to the same proportional depth cut under the same protocol, tolerate it just as well as span_tiny does, with no significant difference in any of the ten backbone–architecture combinations tested.

1.4 Paper organization

Section 2 reviews related work in ear recognition, efficient super-resolution, SR for downstream recognition, and knowledge distillation. Section 3 describes the span_tiny architecture, training recipes, and statistical methodology. Section 4 details the experimental setup across both datasets. Section 5 reports the main and contextual results, a set of three additional mechanisms tested for further improvement and found negative or null (Section 5.7), and three further questions about span_tiny's specific design choices, all now resolved: a completed attention-parameterization ablation, a completed block-removal position ablation, and a resolved-by-inspection recognition-embedding-distillation question (Section 5.8). Section 6 discusses the findings, Section 7 states limitations, and Section 8 concludes.

2 Related Work

2.1 Ear recognition

Ear recognition moved from handcrafted descriptors toward deep-learning approaches over the past decade. Alshazly et al. [11] trained an ensemble of CNNs for unconstrained ear recognition and reported that data augmentation and transfer learning were essential given the small size of available ear datasets. This constraint still shapes experimental design in this area, including the dataset choices made in this paper (Section 4.1). Interest in ear biometrics grew further once widespread mask-wearing made face-based identification unreliable in many settings. Mehta and Singh [12] developed a lightweight, augmentation-based CNN for 2D ear recognition explicitly motivated by this scenario, targeting deployment on CPU-only embedded hardware. Lendering et al. [13] proposed EdgeEar, a hybrid CNN–transformer architecture with low-rank linear layers that cuts parameter count by roughly 50× relative to prior state-of-the-art ear recognition models while achieving the lowest equal-error rate on the UERC2023 benchmark. This shows that the push toward efficient, edge-deployable models seen in SR (Section 2.2) has a direct counterpart on the recognition side of the pipeline studied here. Benzaoui et al. [14] provide a broader survey of 2D, 3D, and hybrid ear recognition approaches, databases, and open challenges.

Publicly available ear datasets vary substantially in scale, acquisition conditions, and subject count. EarVN1.0 [9] is a large-scale, in-the-wild dataset of 28,412 color images from 164 Vietnamese subjects (98 male, 66 female), collected under unconstrained illumination and pose. The Annotated Web Ears (AWE) dataset [10] and its extensions (collectively AWEx in this work) were assembled by the University of Ljubljana ear-recognition research group, the same group behind EdgeEar [13], from web-crawled images of public figures. The base AWE set contains 1,000 images from 100 subjects. This paper extends it with the CVL Ear dataset (16 subjects, 804 images) and an additional "New Images" collection (220 subjects, 2,200 images) from the same research group, giving a combined pool of 336 subjects and 4,004 images (Section 4.1).

2.2 Lightweight/efficient super-resolution

Recent efficient SR research has been substantially driven by the NTIRE efficient super-resolution challenge series. RLFN [2] simplifies feature aggregation to three convolutional layers per residual block, winning the runtime track of NTIRE 2022. ECBSR [3] proposes edge-oriented convolution blocks that re-parameterize multiple operations into a single 3×3 convolution at inference time, targeting real-time performance on commodity mobile SoCs. SAFMN [4] introduces a spatially-adaptive feature modulation mechanism that captures non-local dependencies without the computational cost of dot-product self-attention, achieving parameter counts roughly 3× smaller than IMDN at comparable quality. SMFANet [5] combines a self-modulation feature aggregation module with a partial-convolution feed-forward network to balance local and non-local feature interaction efficiently. SPAN [1], the architecture compressed in this work, replaces learned attention with a parameter-free mechanism based on symmetric activation functions applied to convolutional outputs, winning both the overall-performance and runtime tracks of NTIRE 2024. None of these architectures has been evaluated for depth compression below its original design point under a downstream recognition objective at extremely low input resolution ($\approx 20\times 20$ px).

2.3 Super-resolution for downstream recognition

A separate line of work optimizes SR explicitly for downstream recognition accuracy instead of pixel fidelity. Ataer-Cansizoglu et al. [6] trained an identity-preserving face SR network to minimize distance in a pre-trained face-recognition feature space instead of pixel space, and found that this recognition-aware objective outperformed conventional pixel-fidelity SR for very-low-resolution face verification. The same question has been asked for iris biometrics. Nguyen et al. [7] showed that SR tailored for recognition at a distance improves iris matching more than generic SR does. Ribeiro et al. [8] asked directly whether photo-realistic reconstruction is necessary for iris recognition, and found that the two are related but not the same objective, matching the finding Ataer-Cansizoglu et al. report for faces. The qualitative results in this paper echo the same pattern (Figure 2b, Section 5.1): a near-tied, relatively high PSNR turns out to be driven by accurate reconstruction of hair and background rather than of the ear itself. Ear biometrics has not yet been examined through this lens. The evaluation of span_tiny directly on downstream identity accuracy, rather than on PSNR/SSIM alone, extends the same question to a modality where it has not previously been asked.

2.4 Knowledge distillation in super-resolution

Teacher–student distillation [17] is a common strategy for training compact SR networks, typically combining a pixel-reconstruction loss with a feature- or output-level distillation loss against a larger teacher network. Lee et al. [15] distilled intermediate decoder features from a teacher trained with privileged access to the HR image, letting the student recover high-frequency detail it could not learn from LR input alone. Zhang et al. [16] argued that a core assumption behind most SR distillation methods does not hold: because a teacher's own SR output is itself only a noisy approximation of the true HR image, naively matching student outputs to teacher outputs caps how much the student can improve. Their proposed data-upcycling scheme was designed specifically to work around this ceiling. This observation offers a useful lens on the findings reported here: the loss-ablation in Section 5.3 finds no statistically significant contribution of the distillation term to downstream accuracy at this compression level, which is consistent with distillation gains for SR being real but small and hard to detect against a noisy teacher signal, rather than with distillation being useless. The recipe used in this paper (Section 3.3) is simpler than either of these, and this paper additionally reports a rigorous ablation of its contribution to *downstream recognition accuracy* specifically, a question this literature has not directly addressed.

2.5 Positioning

To our knowledge, no prior work combines further depth compression of a state-of-the-art parameter-free-attention SR architecture, evaluation at an extremely low input resolution (20×20 px) substantially below the design regime of the compared architectures, and multi-seed, multi-backbone statistical validation against a downstream ear-recognition task, confirmed across two independent datasets. This paper addresses that combination.

3 Methodology

3.1 Problem formulation

Given a high-resolution ear crop, a low-resolution counterpart is generated via bicubic downsampling at a fixed scale factor $\times4$ (HR $80\times80 \rightarrow$ LR 20×20 on EarVN1.0; HR $76\times76 \rightarrow$ LR 19×19 on AWEx, since `hr_size` is auto-selected per dataset, Section 4.1). An SR model upsamples the LR image back to the dataset's HR size, and the resulting image is passed to a recognition backbone trained to predict subject identity and, secondarily, gender. Three domains are compared throughout: `lr` (no SR, the LR image bicubically upsampled and fed directly to the recognition backbone, establishing the no-SR baseline), `span_baseline` (the uncompressed SPAN architecture, referred to as `sr_baseline` in project logs), and `span_tiny` (the depth-compressed variant proposed here, referred to as `sr_improved`).

3.2 span_tiny architecture

span_tiny is a depth-compressed variant of SPAN [1]. Where the uncompressed baseline uses `n_blocks=6` Swift Parameter-free Attention Blocks (SPABs) at `feat=48` channels, span_tiny uses `n_blocks=3` at the same channel width, reducing deploy-mode parameters from 0.4263M to 0.230M (−46.0%). Each SPAB (Figure 1a) passes its input x through three sequential 3×3 convolutions, the first two followed by GELU, to obtain raw features h. These raw features feed two parallel, independently-computed paths: a residual sum $u = x + h$, and a parameter-free attention map $v = \mathrm{sigmoid}(h) - 0.5$, a fixed activation symmetric about the origin with no learned parameters. The block output is formed by element-wise multiplication applied last, output $= u \otimes v$: the residual is added before, not after, the attention modulation. Each SPAN stack additionally carries a block-level global residual, in which the feature map immediately after the initial shallow feature-extraction convolution is added back after the final feature-aggregation convolution, immediately before pixel-shuffle upsampling (Figure 1b,c). Halving the number of SPABs removes attention *computation* across the removed blocks but not attention *parameters*, since the mechanism has none. This property is proposed in Section 6 as a partial explanation for why the ablation in Section 5.2 shows minimal accuracy degradation from depth reduction.

Before applying span_tiny to the ear-recognition task, the re-implementation was validated against the original authors' published results. On the SPAN-S benchmark configuration reported in [1] (Table 1), the re-implementation reproduces 99.93% of the reported PSNR, indicating that observed differences in the ear-recognition experiments reflect the compression itself rather than an implementation discrepancy.

Methodological note: re-implementation vs. official architecture. span_tiny and span_large (the uncompressed-depth counterpart used in the depth ablation, Section 5.2) are both instantiated from the same custom `SPAN` class shown in Figure 1a, using plain 3×3 convolutions and differing only in `n_blocks` (3 vs. 6). This makes the span_tiny-vs-span_large comparison (Table 3) a clean, single-variable isolation of depth. By contrast, `span_baseline` (used in Table 2 as the uncompressed reference and referred to as `span_official` in project logs) is instantiated from the original authors' SPAN codebase, which internally uses a reparameterizable Conv3XC convolution design rather than plain 3×3 convolutions. Both implementations are architecturally equivalent at deploy time, since Conv3XC collapses to an equivalent single convolution after re-parameterization, and both are validated against the same published SPAN-S benchmark numbers, but they are not byte-for-byte the same low-level implementation. This is the source of the small parameter-count difference between span_large (0.417M, plain-conv 6-block) and span_baseline (0.4263M, Conv3XC-based 6-block) visible in Table 5, despite both being nominally 6-block, 48-channel configurations. Table 3 (span_tiny vs. span_large) is therefore treated as the methodologically cleaner test of the depth-compression hypothesis, while Table 2 (span_tiny vs. span_baseline) answers the closely related but distinct practical question of how the compressed model compares against deploying the official pre-trained baseline. This distinction is made explicit here so that neither table is over-interpreted as isolating a variable it does not fully isolate.

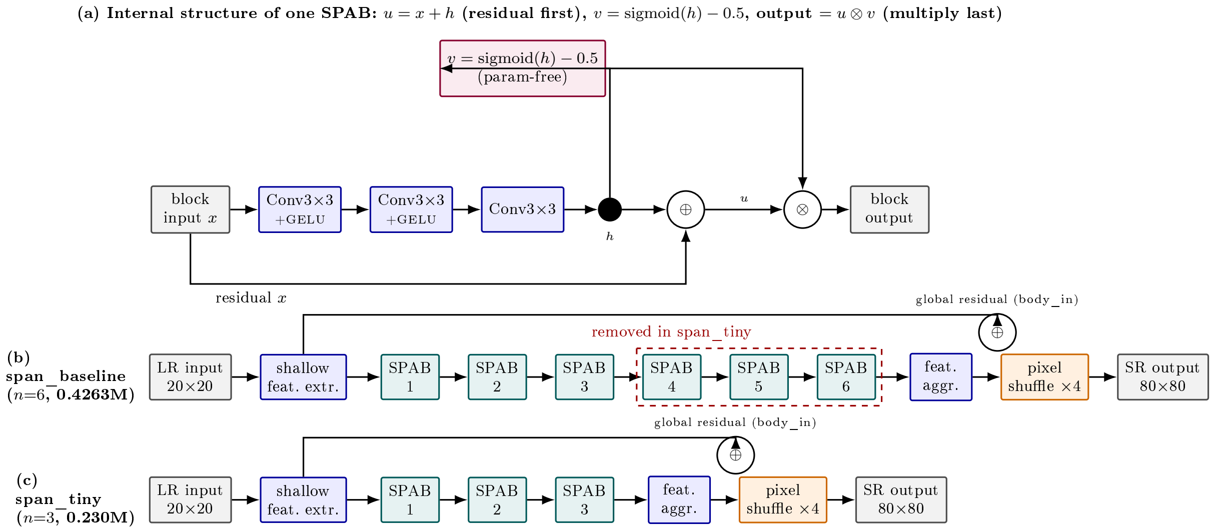


Figure 1: SPAN architecture and depth compression, verified directly against the implementation. (a) Internal structure of a single Swift Parameter-free Attention Block (SPAB): three convolutions (first two followed by GELU) produce raw features h; h feeds two parallel paths, a residual sum $u = x + h$ and a parameter-free attention map $v = \text{sigmoid}(h) - 0.5$, which are combined by *element-wise multiplication last*, output $= u \otimes v$ (residual add precedes, not follows, the attention multiply). Each SPAB stack also carries a block-level global residual (labelled "global residual (body_in)"): the feature map right after shallow feature extraction is added back after the final feature-aggregation convolution, before pixel-shuffle upsampling. (b) span_baseline (via the official SPAN implementation, `span_official`) chains 6 SPABs. (c) span_tiny removes the last 3 SPABs, reducing deploy-mode parameters by 46.0% (0.4263M → 0.230M) while keeping every other component identical. Note: span_tiny and span_large (Section 5.2) are both built from the same custom `SPAN` re-implementation class shown here (plain 3×3 convolutions), differing only in `n_blocks`; `span_baseline`/`span_official` uses the original authors' reparameterizable Conv3XC design internally, which is why its deploy-mode parameter count (0.4263M) differs slightly from span_large's (0.417M) despite both nominally being 6-block, 48-channel configurations – see the methodological note in Section 3.2.

3.3 Training recipe: Track A vs. Track B

Two training tracks are used throughout:

- **Track A** (pixel-loss only, L1 reconstruction loss against the HR ground truth): used exclusively to cross-check SR-quality numbers against values reported in the SR literature. Track A results are **not** used for any comparative claim between architectures.

- **Track B** (distillation recipe, used for all comparative claims): a pre-trained, well-converged `span_official` (the original authors' SPAN implementation) serves as teacher, with loss weights $\lambda_{\text{pixel}} = 1.0$, $\lambda_{\text{distill}} = 1.0$, $\lambda_{\text{identity}} = 0.0$. All primary comparisons in this paper (Sections 5.1, 5.2, 5.4, 5.6) use Track B models, trained under an identical recipe, ensuring a fair comparison across architectures.

The choice of $\lambda_{\text{identity}} = 0.0$ was confirmed by a hyperparameter sweep: $\lambda_{\text{identity}} = 0.3$ produced a statistically significant *negative* effect on downstream accuracy (Cohen's $d = -10.09$, Bonferroni-corrected $p = 0.0163$), supporting the decision to exclude the identity-loss term from the main recipe.

Full hyperparameter and implementation details are given in Section 4.4.

3.4 Recognition backbones

Downstream identity recognition is evaluated on five backbone architectures chosen to span common families of lightweight recognition networks likely to be deployed alongside an SR preprocessing step on resource-constrained devices: EfficientNet-B0 [24], GhostNet-100 [25], MobileNetV2 [22], MobileNetV3-Small [23], and ResNet-18 [21].

3.5 Statistical methodology

For every comparison between two SR domains (e.g., span_tiny vs. lr), downstream recognition accuracy is measured across $n = 5$ random seeds (42, 123, 2024, 44, 999), holding the SR model itself fixed. SR models are trained once per configuration, at a fixed seed of 42; only the downstream recognition training is repeated across seeds to estimate variance, since SR training is deterministic and computationally expensive enough that multi-seed SR training was judged infeasible at the scale of this study. The paired difference in mean accuracy is reported alongside a paired-sample t-test (`scipy.stats.ttest_rel`) and a paired effect size (Cohen's $d_z = \text{mean(difference)}/\text{std(difference)}$) [26]. Because multiple backbones and multiple pairwise comparisons are tested within each result table, a Bonferroni correction [28] is applied across the comparisons within each family. $p_{\text{Bonf}} < 0.05$ is adopted as the threshold for a "statistically significant" claim, with $0.05 \leq p_{\text{Bonf}} < 0.10$ distinguished as "suggestive of a trend" and never described as significant in this paper.

3.6 Data and training integrity validation

During experimentation on the second dataset (AWEx), two training-pipeline issues were identified and corrected by inspecting raw training logs rather than relying solely on aggregated summary statistics, a practice adopted throughout this study after aggregated CSV outputs were found to occasionally mask anomalies visible only in per-epoch logs. First, the `span_official` teacher model exhibited NaN losses during from-scratch training on AWEx (best validation L1 = 0.19, versus ≈0.02–0.03 for healthy runs), traced to numerical overflow under mixed-precision (FP16) training combined with the wrapper's internal `img_range=255` scaling. Disabling automatic mixed precision specifically for this architecture resolved the issue, and PSNR recovered from 9.08 dB to 25.11 dB. Second, a stale teacher checkpoint was inadvertently retained in an auto-generated fine-tuning configuration, causing an initial transfer-learning run to regress relative to its zero-shot starting point. Retraining after the teacher-checkpoint issue was fixed corrected this: fine-tuned PSNR reached 25.135 dB, above the zero-shot value of 24.989 dB, in the expected direction. All results reported in Section 5.6 use the corrected, post-fix data. This validation process is described here both for transparency and because it may offer methodological guidance for others training SPAN-family architectures under mixed precision.

4 Experimental Setup

4.1 Datasets

EarVN1.0 (primary dataset) [9]: 164 subjects, used with an automatically-selected low-resolution generation threshold (`hr_source_min=41`, the 20th percentile of the dataset's native resolution distribution). LR images are generated by bicubic downsampling from an 80×80 HR crop to 20×20 (scale ×4).

AWEx (secondary dataset, generalization check): 336 subjects, 4,004 images, assembled from three sources released by the same research group (University of Ljubljana): the original AWE set (100 subjects, 1,000 images), the CVL Ear dataset (16 subjects, 804 images), and an additional "New Images" collection (220 subjects, 2,200 images) [10]. All 336 subjects are pooled into a single set with a custom per-subject train/validation/test split (2,260/408/572 images), rather than AWEx's official open-set split (train = AWE+CVL, test = New Images, with disjoint subject identities). The official split is designed for open-set verification, whereas this study addresses closed-set identity classification, matching the protocol used for EarVN1.0. `hr_source_min` is likewise selected automatically (38, the 20th percentile of AWEx's native resolution distribution); the different numeric value relative to EarVN1.0 reflects the two datasets' differing native resolution distributions under an identical selection algorithm, not a protocol inconsistency.

The unmodified AWE dataset (100 subjects, exactly 10 images/subject, ≈5.7 images/subject after splitting) was initially used as the generalization check. Inter-seed variance in downstream accuracy proved comparable in magnitude to the mean effect sizes under study, precluding reliable conclusions even after applying transfer learning and backbone-freezing to reduce variance. The larger, pooled AWEx set was adopted instead. This substitution is reported here as a methodological decision driven by statistical power, not as a post hoc search for a more favorable dataset.

Table 1 reports native-resolution percentiles (short side, pixels) and split sizes for both datasets, measured directly on the full raw image sets before thresholding.

Table 1: Descriptive statistics for both datasets (native short-side resolution percentiles and split sizes), measured on the full raw image sets.

Statistic	EarVN1.0 (28,412 images)	AWEx (4,004 images)
Native short side: min / p5 / p10	13 / 27 / 32 px	12 / 27 / 31 px
Native short side: p20 / p25 / p50	41 / 46 / 77 px	38 / 41 / 60 px
Native short side: p75 / p90 / max	122 / 161 / 472 px	95 / 149 / 557 px
`hr_source_min` (p20 threshold)	41 px (80.85% kept)	38 px (80.92% kept)
`hr_size` / LR size (auto-selected, ×4)	80 / 20 px	76 / 19 px
`too_small_max` (real-LR holdout)	<20 px (173 images)	<20 px (32 images)
Train / validation / test images	16,077 / 3,415 / 3,480	2,260 / 408 / 572
Subjects (identities)	164 (all represented in every split)	336 (all represented in every spl
Gender distribution (subjects)	98 male / 66 female	not available (see note below)

Gender labels for EarVN1.0 follow the folder-range convention documented by the dataset's original authors (subjects 1–98 male, 99–164 female) [9], and the resulting 98/66 split above is computed directly from `splits.json` (consistent across all three splits: 62.1%/37.9% in train, 62.0%/38.0% in val, 62.1%/37.9% in test). AWEx carries no equivalent ground truth: it is assembled from three source collections whose subjects are renumbered by directory-listing order during preprocessing (Section 4.1), an ordering with no relationship to gender, so no gender distribution or gender-classification result is reported for AWEx anywhere in this paper.

4.2 Evaluation protocol

SR image quality is evaluated with PSNR and SSIM [19] computed on the ear region of interest (ROI) only, and with LPIPS [20] (using ImageNet-normalized features), against a consistent bicubic-upsampling protocol. Downstream recognition performance is evaluated via rank-1 identity accuracy (the primary metric reported throughout this paper), rank-5 accuracy, gender-classification accuracy, and AUC/EER, with full confusion matrices retained for error analysis.

4.3 Baselines

The following SR configurations are compared: `edsr_teacher` (an EDSR-based [18] upper-bound reference), `span_baseline`/`span_official` (uncompressed SPAN), `span_large` (depth-ablation counterpart to span_tiny), and, in the contextual comparison of Sections 5.4/5.6, `rlfn`,

rlfn_adapted, ecbsr, safmn, and smfanet, each trained under both Track A and Track B where applicable. rlfn_adapted uses a modified ESA-pooling kernel/stride configuration (kernel=3, stride=2) rather than the original authors' setting (kernel=7, stride=3), because at 20×20 input resolution the original configuration was found to degenerate toward a near-channel-attention mechanism. rlfn_adapted is therefore used as the primary RLFN baseline throughout, with the unmodified rlfn reported only as a secondary reference.

4.4 Implementation details

All models are trained with Adam [27] at a fixed learning rate (no scheduler), relying on early stopping (patience 10 epochs on the relevant validation metric, maximum 500 epochs) rather than a decay schedule to control convergence. Batch size is 64 throughout. Learning rates differ by training stage: 1×10^{-3} for the uncompressed SPAN teacher and other from-scratch Track-A/Track-B SR baselines; 1×10^{-4} for span_tiny's distillation recipe and for learned block pruning (with a further 10× reduction for the post-harden fine-tuning stage, Section 5.7); 3×10^{-4} (weight decay 5×10^{-4}) for all recognition-backbone training. Automatic mixed precision is used for recognition training on CUDA, but deliberately *not* for SR training, after the FP16 numerical-overflow issue described in Section 3.6. Data augmentation is deliberately minimal: LR-domain images are bicubically resized to the target resolution (a no-op for the hr/sr_baseline/sr_improved domains, whose images are already exported at the correct size); no random cropping, color jitter, or flipping is applied. A random-horizontal-flip augmentation used in earlier development was deliberately removed, since flipping an ear image can turn a left-ear morphology into one resembling a right ear – ears, unlike faces, are not left-right symmetric, making this augmentation biometrically invalid for this modality rather than a free source of extra training diversity.

All experiments were run with PyTorch 2.11.0 (CUDA 13.0, cuDNN 9.19), torchvision 0.26.0, and timm 1.0.26, on a single NVIDIA GeForce RTX 3080 (10 GB, compute capability 8.6).

5 Results

5.1 Main result (EarVN1.0): accuracy ordering across five recognition backbones

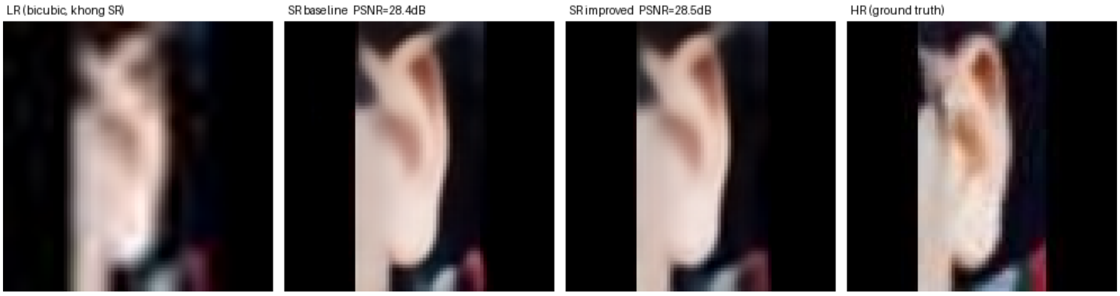
Table 2 reports downstream rank-1 identity accuracy for the lr (no-SR), span_tiny, and span_baseline domains across all five recognition backbones, together with paired statistical tests for the two comparisons central to this paper's claim.

Table 2: Downstream identity accuracy (rank-1) on EarVN1.0, $n = 5$ seeds. "tiny>lr" and "tiny vs. baseline" report paired Cohen's d and Bonferroni-corrected p-values (family of 3 comparisons per backbone: lr–baseline, lr–tiny, baseline–tiny).

Backbone	lr	span_tiny	span_baseline	tiny > lr (d, p_{Bonf})	tiny vs. baseline (d, p_{Bonf})
EfficientNet-B0	0.5125	0.5485	0.5604	d=8.38, p=0.0001 (sig.)	d=−2.45, p=0.0163 (sig., tiny lower)
GhostNet-100	0.4675	0.5115	0.5205	d=9.10, p=0.0001 (sig.)	d=−1.36, p=0.1163 (n.s.)
MobileNetV2	0.4444	0.5026	0.5125	d=7.75, p=0.0002 (sig.)	d=−0.74, p=0.5135 (n.s.)
MobileNetV3-Small	0.4002	0.4423	0.4398	d=4.41, p=0.0018 (sig.)	d=0.27, p=1.0 (n.s., tiny higher)
ResNet-18	0.4575	0.4825	0.4882	d=2.20, p=0.0239 (sig.)	d=−0.66, p=0.6379 (n.s.)

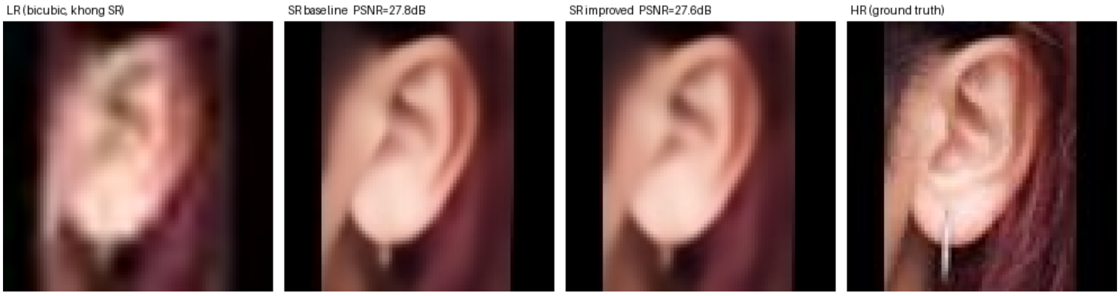
span_tiny > lr reaches statistical significance ($p_{\text{Bonf}} < 0.05$) in 5/5 backbones, with large effect sizes throughout (Cohen's d 2.2–9.1). span_baseline > lr is equally strong and significant in 5/5 backbones (d 3.8–10.6, not tabulated separately). The accuracy cost of compression itself, span_tiny vs. span_baseline, reaches significance in only 1/5 backbones (EfficientNet-B0, a 1.2-percentage-point cost); the other four are statistically indistinguishable, and MobileNetV3-Small shows span_tiny fractionally *higher* than the baseline (not significant). This is a stronger and cleaner result than a coarser reading of the data might suggest: the benefit of using SR at all is large and universal, while the specific cost of halving SPAN's depth is small and, in 4/5 backbones, not statistically detectable.

Figure 2 shows qualitative examples on real EarVN1.0 test images. Across the full 20-sample export, the per-image PSNR gap (span_baseline − span_tiny) ranges from −0.062 dB, with span_tiny marginally ahead (Figure 2a), to +2.645 dB, the largest gap observed (Figure 2e). Samples (a)–(e) are selected to span this full range rather than to favor either model. Two patterns are visible. First, the visual gap between span_baseline and span_tiny tracks the PSNR gap: it is imperceptible when the two are close (Figure 2a–b) and visible mainly as reduced fine-structure sharpness in the antihelix/concha region when the gap is largest (Figure 2e), consistent with a small, non-catastrophic accuracy cost rather than a qualitatively different failure mode. Second, Figure 2b illustrates a limitation of pixel-fidelity metrics noted in Section 2: the ear itself is largely occluded by hair in this sample, so the near-tied, relatively high PSNR (30.92 vs. 30.89 dB) is driven substantially by accurate reconstruction of the large, low-detail hair and background regions rather than by ear-specific structure. PSNR and downstream identity accuracy are related but not interchangeable signals.

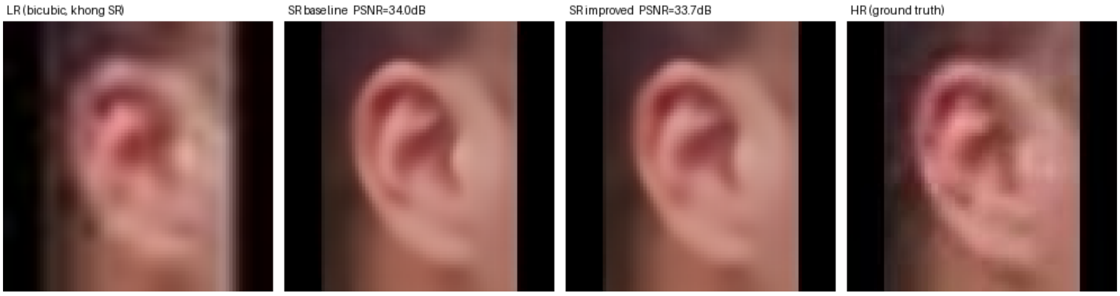


(a) baseline 28.654 dB/SSIM 0.8968, span_tiny 28.716 dB/SSIM 0.8824 – span_tiny marginally ahead on PSNR

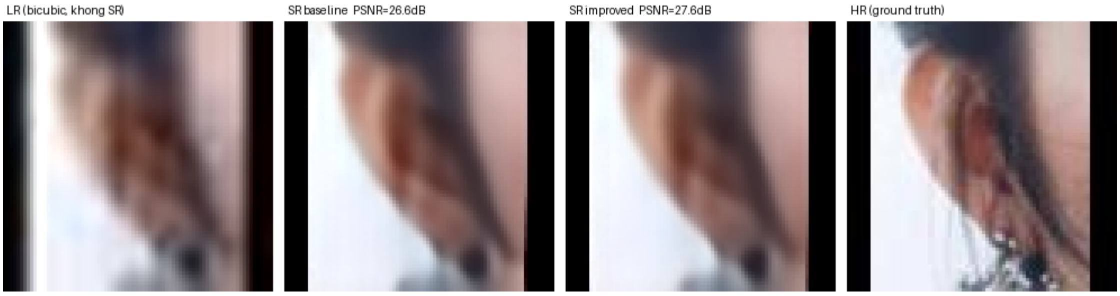
(b) baseline 30.920 dB/SSIM 0.8580, span_tiny 30.890 dB/SSIM 0.8544 – near-tie, ear largely occluded by hair



(c) baseline 25.656 dB/SSIM 0.8576, span_tiny 25.252 dB/SSIM 0.8395 – lowest-PSNR pair in the 20-sample export



(d) baseline 34.946 dB/SSIM 0.9067, span_tiny 33.716 dB/SSIM 0.8908 – highest-PSNR pair in the 20-sample export



(e) baseline 31.413 dB/SSIM 0.9075, span_tiny 28.768 dB/SSIM 0.8813 – largest observed gap across all 20 exported EarVN1.0 samples

Figure 2: Qualitative comparison on five EarVN1.0 test images, each showing (left to right) the bicubic-upsampled LR input (no SR), span_baseline output, span_tiny output, and the HR ground truth. Samples (a)–(e) were selected from a 20-sample export to span the full observed per-image PSNR gap, from span_tiny marginally ahead (a) to the largest gap in favor

These per-image values are computed over the full image frame, including any letterbox padding, whereas the dataset-level averages in Table 5 are computed on the ear region of interest only (Section 4). The two are therefore not directly comparable in absolute terms; the same distinction applies to the AWEx qualitative comparison (Figure 3).

5.2 Ablation: depth (span_tiny vs. span_large)

Table 3: span_tiny vs. span_large, downstream identity accuracy, EarVN1.0, $n = 5$ seeds. Difference and d are span_large − span_tiny; p_{Bonf} corrected across the 6-comparison family reported by this ablation's own aggregation (Section 3.5).

Backbone	span_tiny	span_large	Difference	Cohen's d	p_{Bonf}
EfficientNet-B0	0.5485	0.5568	+0.0083	3.17	0.0125 (sig.)
GhostNet-100	0.5115	0.5140	+0.0025	1.33	0.244
MobileNetV2	0.5026	0.5044	+0.0018	0.29	1.0
MobileNetV3-Small	0.4423	0.4473	+0.0050	0.37	1.0
ResNet-18	0.4825	0.4863	+0.0038	0.45	1.0

4/5 backbones show no statistically significant difference between span_tiny (0.230M parameters) and span_large (0.417M parameters). The fifth, EfficientNet-B0, shows a small but statistically significant cost (p_{Bonf}=0.0125), span_large ahead by 0.83 percentage points. This is still the strongest single piece of evidence for the paper's central claim, precisely stated: reducing depth from 6 to 3 SPABs is accuracy-neutral in the large majority of backbones tested and, where it is not, the cost is small (under 1 percentage point) – not a claim of zero cost in every setting. On AWEx (Section 5.6), the same comparison is statistically non-significant in 5/5 backbones, though at a smaller sample ($n = 3$ seeds; see Section 7).

5.3 Ablation: loss components

Table 4: Downstream accuracy, all pairwise comparisons among four training-loss configurations (MobileNetV2, EarVN1.0, $n = 5$ seeds). Difference is the row label's second configuration minus its first; p_{Bonf} corrected across all 6 comparisons.

Comparison	Difference	Cohen's d	p_{Bonf}	Verdict
pixel+distill+identity vs. pixel+identity	−0.0072	−0.84	1.0	n.s.
pixel+distill+identity vs. pixel+distill	+0.0155	1.93	0.0746	trend (identity term costs acc.)
pixel+distill+identity vs. pixel-only	+0.0119	3.13	0.0131	**sig. (identity term costs acc.)**
pixel+identity vs. pixel+distill	+0.0228	2.85	0.0187	**sig. (identity term costs acc.)**
pixel+identity vs. pixel-only	+0.0191	2.61	0.0257	**sig. (identity term costs acc.)**
pixel+distill vs. pixel-only	−0.0037	−0.40	1.0	n.s.

The six configurations resolve into two statistically-tied clusters: {pixel+distill, pixel-only} both significantly or near-significantly outperform {pixel+distill+identity, pixel+identity} (3/4 cross-cluster comparisons significant, 1/4 at trend level), while the two within-cluster comparisons are both non-significant. In other words, whichever configuration includes the identity-loss term scores lower, regardless of whether distillation is also present, and this reaches significance in the majority of the relevant comparisons. This corroborates, on an independent ablation design, the standalone $\lambda_{\text{identity}}$ sweep already reported in Section 3.3 (d=−10.09 at $\lambda_{\text{identity}}$=0.3): the identity-loss term reliably hurts downstream accuracy at this compression level. By contrast, the distillation term's own marginal contribution – pixel+distill vs. pixel-only – remains statistically indistinguishable (d=−0.40, p_{Bonf}=1.0); this specific negative result is unchanged from earlier analysis and is reported as such, without reinterpretation.

5.4 Contextual comparison with other lightweight SR architectures

Table 5 reports SR image-quality metrics (single run, $n = 1$) for all evaluated architectures. Table 6 reports downstream accuracy comparisons against span_tiny under the shared Track B distillation recipe (MobileNetV2, $n = 5$ seeds).

Table 5: SR image quality on EarVN1.0 (ROI-only, ×4, $n = 1$).

Model	PSNR (dB)	SSIM	LPIPS↓	Params (M, deploy)	Latency (ms)
EDSR (teacher)	27.434	0.8307	0.1256	4.9292	0.895
span_baseline	27.232	0.8261	0.1409	0.4263	0.566
span_tiny	**27.053**	**0.8193**	**0.1576**	**0.230**	**0.256**
span_large	27.137	0.8228	0.1529	0.417	0.464
RLFN (Track A, unmodified, secondary)	27.210	0.8245	0.1379	0.3278	0.684
RLFN-adapted (Track A)	27.192	0.8251	0.1335	0.3278	0.681
ECBSR (Track A)	26.074	0.7883	0.1722	0.0167	0.137
SAFMN (Track A)	27.355	0.8287	0.1350	0.2395	1.824
SMFANet (Track A)	20.906	0.6038	0.3104	0.0844	1.851
RLFN-adapted (Track B, distilled)	27.111	0.8211	0.1522	0.3278	0.710
ECBSR (Track B, distilled)	25.847	0.7813	0.1899	0.0167	0.135
SAFMN (Track B, distilled)	27.141	0.8218	0.1559	0.2395	1.799
SMFANet (Track B, distilled)	26.289	0.7953	0.1833	0.0844	1.878
SAFMN half-depth (Track B, distilled)	27.002	0.8176	0.1564	0.1281	1.103
SMFANet half-depth (Track B, distilled)	26.140	0.7900	0.1935	0.0477	1.125

span_tiny is the second-fastest model, behind only ECBSR, and third-smallest by parameter count, behind ECBSR and SMFANet; it likewise ranks third-lowest in PSNR, ahead of only SMFANet and ECBSR. This is consistent with a framing around efficiency rather than raw image-quality superiority – but, as Table 6 shows next, this ranking does not carry over to downstream recognition accuracy, the metric this paper is actually organized around. SMFANet's pixel-loss-only (Track A) result (20.906 dB) was investigated with a dedicated debugging script. The checkpoint itself was confirmed healthy, and the low score reflects SMFANet's difficulty converging under pure pixel supervision at 20×20 resolution. With distillation supervision (Track B), SMFANet recovers to 26.289 dB (+5.4 dB), and Track B results are therefore used for all comparative claims involving SMFANet. The half-depth SAFMN/SMFANet rows support the attention-parameterization ablation (Section 5.8).

Table 6: span_tiny vs. four other Track-B lightweight SR architectures, downstream accuracy, EarVN1.0, all five recognition backbones, $n = 5$ seeds. d is span_tiny minus the other architecture (positive = span_tiny ahead); p_{Bonf} corrected across the 6-comparison family reported by each backbone's own aggregation (Section 3.5).

Backbone	vs. RLFN-adapted	vs. SAFMN	vs. ECBSR	vs. SMFANet
EfficientNet-B0	d=−0.79, n.s.	d=−0.89, n.s.	d=+3.37, **sig.**	d=+6.97, **sig.**
GhostNet-100	d=−0.98, n.s.	d=+0.24, n.s. (tiny ahead)	d=+2.94, **sig.**	d=+2.30, **sig.**
MobileNetV2	d=−0.17, n.s.	d=−0.24, n.s.	d=+3.79, **sig.**	d=+2.08, trend
MobileNetV3-Small	d=−0.22, n.s.	d=+0.11, n.s. (tiny ahead)	d=+2.41, **sig.**	d=+1.37, n.s. (tiny ahead)
ResNet-18	d=−0.56, n.s.	d=+0.18, n.s. (tiny ahead)	d=+1.85, trend	d=+1.49, n.s. (tiny ahead)

Across all five backbones, span_tiny is never significantly beaten by any of the four architectures compared. Against RLFN-adapted and SAFMN, the two architectures that led on raw PSNR (Table 5), span_tiny is statistically tied in 5/5 backbones, and is nominally ahead (though not significantly) in 3/5 backbones against SAFMN. Against ECBSR, span_tiny wins with significance in 4/5 backbones and at trend level in the fifth (ResNet-18, p_{Bonf}=0.0862). Against SMFANet, span_tiny wins with significance in 2/5, at trend level in 1/5 (MobileNetV2, p_{Bonf}=0.0577), and is tied (nominally ahead) in the remaining 2/5. This is a materially different picture from raw PSNR/SSIM ranking, where span_tiny placed third-lowest: on the metric the paper is actually organized around, span_tiny is competitive with the best-performing alternatives and significantly outperforms the two weakest ones. This reinforces the point made in Section 2.2 and revisited in the qualitative analysis (Section 5.1): pixel-fidelity metrics are an imperfect proxy for recognition-relevant SR quality at this compression level. This comparison remains contextual to the paper's central claim (Sections 5.1–5.2), which concerns depth compression *within* the SPAN family, not a claim that span_tiny is the best-performing lightweight SR architecture in absolute terms.

5.5 Auxiliary validation: real low-resolution holdout

On a held-out set of 173 genuinely low-resolution ear images (rather than synthetically downsampled ones), downstream identity accuracy *reverses* the main result's ordering: no-SR = 0.1965, span_tiny = 0.1561, span_baseline = 0.1214 (single run, $n = 1$, mobilenet_v2 only, not multi-seed); gender accuracy shows the same reversed ordering (no-SR = 0.8555, span_tiny = 0.8324, span_baseline = 0.8382), which argues against this being sampling noise specific to one task. The likely cause is a resolution mismatch: every one of the 173 holdout images has a native short side strictly below 20 px (range 13–19 px, mean 17.8 px) – below the resolution the SR models were trained to receive as input. The SR models were trained exclusively to invert one specific, clean synthetic degradation (bicubic downsampling from an 80 px HR crop); genuinely-captured images this small do not share that degradation's statistics, and many additionally have non-square aspect ratios (mean long side 25.7 px against a 17.8 px short side) that cause the shared letterbox-resize routine (Section 3.2) to shrink the short side further still and introduce a larger padded region before the SR model sees the image. Under this distribution mismatch, a model trained to confidently reconstruct fine detail from a specific degradation can readily hallucinate plausible-looking but incorrect high-frequency structure, actively harming downstream recognition, while plain bicubic upsampling (the no-SR path) never attempts this reconstruction and therefore cannot introduce it. This is a known failure mode in the SR literature – models trained on synthetic degradation transferring poorly to real-world degradation – and is discussed further as a boundary condition on the paper's central claim in Section 7, rather than treated as evidence against it: the main claim is specifically about the synthetic bicubic-downsampling regime defined in Section 3, and this auxiliary check probes a related but distinct question that the main experiments were never designed to answer.

5.6 Cross-dataset generalization (AWEx)

All AWEx results below were generated after every correction described in Section 3.6 and every statistical-methodology refinement described in Section 3.5 had already been incorporated; no AWEx figure from an earlier, uncorrected stage of this study is reused anywhere in this paper.

5.6.1 From-scratch track

Table 7 reports results from models trained from scratch on AWEx (no transfer of weights or teacher checkpoints from EarVN1.0). p_{Bonf} is corrected across the 3-comparison family per backbone (lr–baseline, lr–tiny, baseline–tiny), matching Table 2's convention.

Table 7: Downstream identity accuracy on AWEx, from-scratch training, $n = 5$ seeds.

Backbone	lr	span_baseline	span_tiny	baseline>lr	tiny>lr	tiny vs. baseline
EfficientNet-B0	0.1290	0.1619	0.1524	sig. (p=0.0066)	sig. (p=0.0127)	n.s. (p=0.1266)
GhostNet-100	0.0892	0.1234	0.1304	sig. (p=0.0129)	sig. (p=0.0014)	n.s. (p=0.2226, tiny higher)
MobileNetV2	0.0885	0.1196	0.1210	sig. (p=0.0293)	trend (p=0.0543)	n.s. (p=1.0, tiny higher)
MobileNetV3-Small	0.0888	0.1115	0.1084	sig. (p=0.0208)	sig. (p=0.0116)	n.s. (p=0.698)
ResNet-18	0.1748	0.1808	0.1860	n.s. (p=0.6719)	n.s. (p=0.543)	n.s. (p=1.0, tiny higher)

An earlier stage of this study, predating the corrections in Section 3.6, had reported ResNet-18's from-scratch baseline scoring *below* the no-SR condition on this track – the only off-trend result across both datasets at that stage. This does not replicate under the corrected setup (Table 7: baseline > lr for ResNet-18, +0.6 percentage points, though not significant at $n = 5$); the earlier result is attributed to the since-corrected issues rather than to a property of the backbone, and is not carried forward as a limitation. `span_tiny` vs. `span_large` on this track shows no statistically significant difference in 5/5 backbones (Section 5.2), further corroborating the depth-compression finding on a second, independent dataset.

Table 8 reports SR image quality for the from-scratch track.

Table 8: SR image quality on AWEx, from-scratch training ($n = 1$).

Model	PSNR (dB)	SSIM	LPIPS↓
EDSR (teacher)	25.584	0.7687	0.1642
span_baseline (span_official)	25.255	0.7607	0.1763
span_tiny	**24.698**	**0.7392**	**0.2385**
span_large	24.435	0.7267	0.2834

All four models score within a coherent 24.4–25.6 dB band with no anomaly, in contrast to an earlier stage of this study in which `span_large` had scored anomalously low (≈19 dB, 5–6 dB below every other architecture) with no corresponding downstream-accuracy effect; that anomaly is not present under the current measurements and is not carried forward as an open item. The contextual comparison against RLFN/ECBSR/SAFMN/SMFANet (Table 5, Table 6) was run on EarVN1.0 only in this phase of the study; extending it to AWEx is noted as a direction for future work (Section 7) rather than reported with placeholder values.

5.6.2 Transfer-learning tracks (primary result for AWEx)

Given AWEx's smaller per-subject sample size, two transfer-learning protocols initialized from EarVN1.0-trained weights are evaluated: full fine-tuning of the entire recognition backbone (Table 9), and a frozen-backbone variant that only fine-tunes the classification head (Table 10). Both are reported as primary results, rather than treating one as a fallback for the other, because they answer a related but distinct question and, as discussed below, do not agree in every respect.

Table 9: Downstream identity accuracy on AWEx, transfer-learning track (full fine-tuning), $n = 5$ seeds. p_{Bonf} corrected across the 3-comparison family per backbone (lr–baseline, lr–tiny, baseline–tiny), matching Table 2's convention; only `tiny vs. baseline`, the comparison central to this paper's claim, is tabulated, with `baseline>lr` and `tiny>lr` discussed in the main text.

Backbone	lr	span_baseline (transfer)	span_tiny (transfer)	tiny vs. baseline
EfficientNet-B0	0.1780	0.2063	0.1864	n.s. (p=0.1613)
GhostNet-100	0.1301	0.1643	0.1475	n.s. (p=0.1207)
MobileNetV2	0.1178	0.1510	0.1402	n.s. (p=0.625)
MobileNetV3-Small	0.1091	0.1416	0.1469	n.s. (p=1.0, tiny higher)
ResNet-18	0.1021	0.1196	0.1070	n.s. (p=0.549)

Under full fine-tuning, `span_baseline` > `lr` reaches significance in 4/5 backbones (all but ResNet-18) and `span_tiny` > `lr` in 2/5 (MobileNetV2, MobileNetV3-Small; d=3.22 and 3.23). Critically for this paper's central claim, `span_tiny` vs. `span_baseline` is non-significant in 5/5 backbones – the compression-safety finding replicates under this protocol at full strength.

Table 10: Downstream identity accuracy on AWEx, transfer-learning track (frozen backbone), $n = 5$ seeds. p_{Bonf} corrected across the 3-comparison family per backbone (lr–baseline, lr–tiny, baseline–tiny), matching Table 2's convention; only `tiny vs. baseline`, the comparison central to this paper's claim, is tabulated, with `baseline>lr` and `tiny>lr` discussed in the main text.

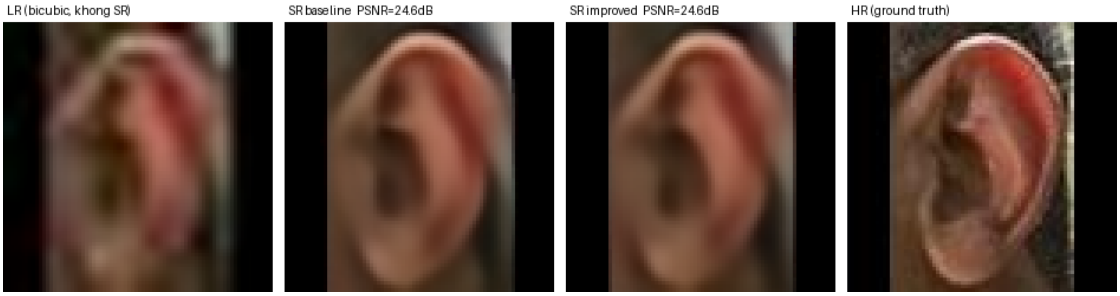
Backbone	lr	span_baseline (transfer)	span_tiny (transfer)	tiny vs. baseline
EfficientNet-B0	0.1706	0.2045	0.2126	n.s. (p=1.0, tiny higher)
GhostNet-100	0.1367	0.1479	0.1647	n.s. (p=0.871, tiny higher)
MobileNetV2	0.1329	0.1609	0.1329	n.s. (p=0.212)
MobileNetV3-Small	0.1052	0.1444	0.1469	n.s. (p=1.0, tiny higher)
ResNet-18	0.1070	0.1154	0.0951	n.s. (p=0.173)

Under the frozen-backbone protocol, `span_baseline` > `lr` reaches significance in 2/5 backbones (MobileNetV2, MobileNetV3-Small) and a trend in a third (EfficientNet-B0, p_{Bonf}=0.0973); `span_tiny` > `lr` reaches significance in 3/5 (EfficientNet-B0, GhostNet-100, MobileNetV3-Small). As with full fine-tuning, `span_tiny` vs. `span_baseline` is non-significant in 5/5 backbones. The compression-safety finding is therefore confirmed under both transfer protocols and the from-scratch track alike: 0/5 backbones show a significant accuracy cost from compression under any of the three protocols tested on AWEx. The two protocols do disagree on secondary detail: under frozen backbone, `span_tiny` is nominally *above* `span_baseline` in 3/5 backbones (never significantly), MobileNetV2 shows tiny exactly tied with `lr`, and ResNet-18 is the one case across all AWEx protocols where `span_tiny` falls nominally below `lr` (not significant, p_{Bonf}=1.0). ResNet-18 is, across all three AWEx protocols, consistently the backbone with the weakest or most inconsistent evidence that SR helps over no-SR at all – always in a non-significant, sometimes reversed direction – while never showing a significant compression cost; this is noted as a backbone-specific limitation (Section 7), not as evidence against the compression-safety claim, which does not depend on this secondary comparison.

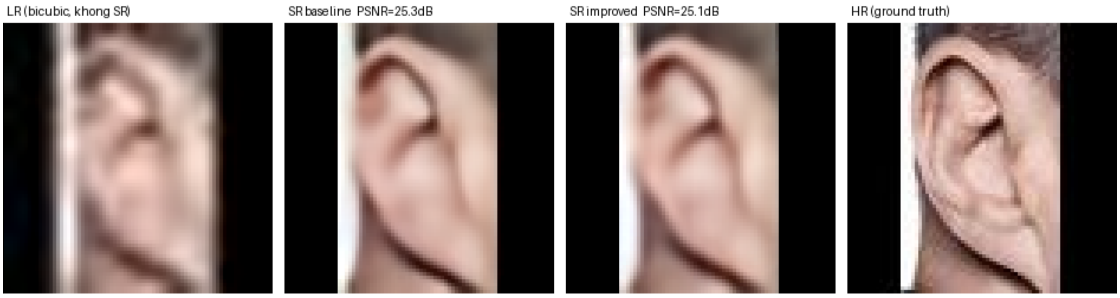
An identical real-low-resolution-holdout check to Section 5.1 was also run on AWEx (mobilenet_v2, frozen-backbone checkpoints, $n = 1$): no-SR = 0.0, span_baseline= 0.0625, span_tiny= 0.0938. Unlike the EarVN1.0 holdout (Section 5.1), both SR variants clearly beat no-SR here rather than reversing it; span_tiny is nominally ahead of span_baseline, echoing the same-direction flip seen for this backbone-adjacent pattern elsewhere in this study (Tables 2, 10). With $n = 1$ and only 32 images (Section 7), this is not treated as contradicting the EarVN1.0

finding, only as insufficient evidence on its own either way.

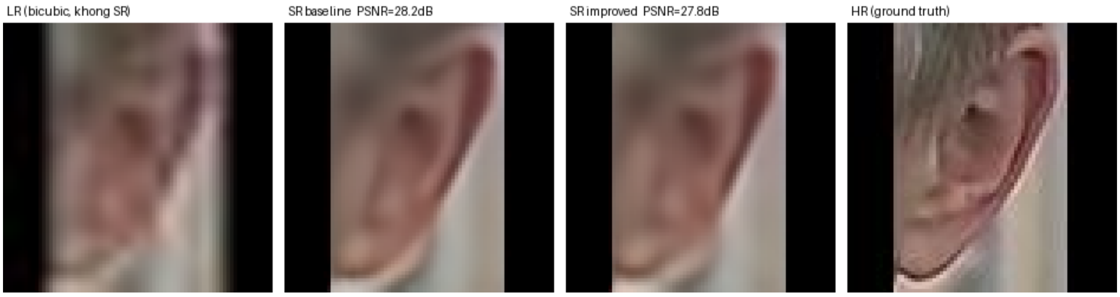
Figure 3 shows qualitative examples on AWEx test images, from the transfer-learning track (full fine-tuning protocol, Table 9), using the SR checkpoints fine-tuned specifically for that protocol. Across the 15 exported AWEx test samples, the per-image PSNR gap (span_baseline − span_tiny) ranges from -0.042 dB (sample index 13, span_tiny fractionally ahead) to $+1.358$ dB (sample index 8, the largest gap observed) – a narrower range than in an earlier export tied to a different (frozen-backbone) checkpoint, consistent with this being a genuinely different SR checkpoint rather than a re-labeling of the same one. The five panels shown are selected by PSNR-gap quartile across the full 15-sample export (minimum, Q1, median, Q3, maximum) rather than by visual inspection, so the selection is not biased toward either model; none of the five is grayscale.



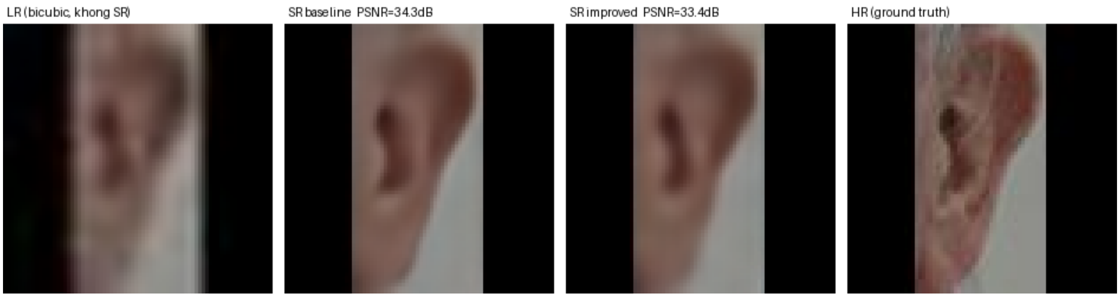
(a) baseline 24.602 dB/SSIM 0.7787, span_tiny 24.644 dB/SSIM 0.7733 – span_tiny marginally ahead, minimum gap observed



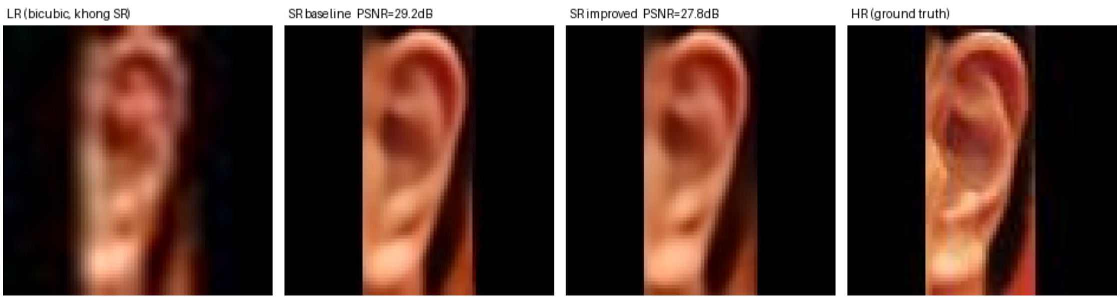
(b) baseline 25.318 dB/SSIM 0.8643, span_tiny 25.092 dB/SSIM 0.8557 – gap 0.226 dB (Q1)



(c) baseline 28.190 dB/SSIM 0.8278, span_tiny 27.827 dB/SSIM 0.8168 – gap 0.363 dB (median)



(d) baseline 34.258 dB/SSIM 0.9302, span_tiny 33.351 dB/SSIM 0.9218 – gap 0.907 dB (Q3)



(e) baseline 29.204 dB/SSIM 0.8962, span_tiny 27.846 dB/SSIM 0.8745 – largest gap observed across all 15 exported AWEx samples

Figure 3: Qualitative comparison on five AWEx test images, each showing (left to right) the bicubic-upsampled LR input (no SR), span_baseline output, span_tiny output, and the HR ground truth. Samples (a)–(e) are selected by PSNR-gap quartile (minimum, Q1, median, Q3, maximum) across the 15-sample export, from span_tiny marginally ahead (a) to the largest gap in favor of span_baseline (e).

Summary of Section 5.6. The paper's central compression-safety claim – `span_tiny` statistically indistinguishable from `span_baseline` – is confirmed on a second, independent dataset under *three* separate protocols (from-scratch, full-fine-tune transfer, frozen-backbone transfer), with 0/5 backbones showing a significant cost under any of them. The secondary question, whether SR helps over no-SR at all, is directionally positive across most backbone×protocol combinations but noisier and protocol-dependent in exactly which backbones reach significance, consistent with AWEx's smaller per-subject training sample (≈6.7 images/subject vs. substantially more on EarVN1.0) rather than a contradiction of the EarVN1.0 findings; this distinction is discussed further in Section 6. ResNet-18 is the one backbone that does not fit this pattern cleanly on AWEx (Section 5.6.2) and is reported as a limitation rather than smoothed over.

5.7 Additional mechanisms tested for further improvement: a negative-result ablation study

Beyond the core block-count reduction, three further mechanisms were designed and tested to determine whether span_tiny's downstream accuracy could be improved further: a richer distillation loss, a saliency-weighted identity-critical loss, and learned/differentiable block pruning as an alternative to span_tiny's fixed, hand-picked block selection. Each was screened at reduced scale first, then validated to the same statistical standard as the main result ($n = 5$ seeds, all five recognition backbones, EarVN1.0) once a plausible configuration was identified. All three are reported here as negative or null results, exactly as measured, following the same reporting standard already applied to the distillation-term ablation in Section 5.3.

5.7.1 Feature-level distillation ("KD v2")

This mechanism adds a feature-level distillation term (matching intermediate SPAN features between student and teacher) to the existing pixel+distillation recipe. A related variant additionally combining this with a multi-judge identity loss (an ensemble of independently-trained recognition backbones providing identity supervision) was screened alongside it but performed no better during screening; the feature-only variant, the stronger candidate, is the one carried forward to the full $n = 5$-seed validation reported here. Table 11 compares it against the main recipe (span_tiny).

Table 11: KD v2 ablation vs. span_tiny's main recipe, downstream accuracy, EarVN1.0, $n = 5$ seeds.

Backbone	span_tiny	span_tiny+KDv2	Cohen's d	p_{Bonf}
EfficientNet-B0	0.5485	0.5512	0.26	1.0
GhostNet-100	0.5115	0.5142	0.30	1.0
MobileNetV2	0.5026	0.4906	−0.89	0.3542
MobileNetV3-Small	0.4423	0.4352	−0.42	1.0
ResNet-18	0.4825	0.4848	0.69	0.5992

No backbone shows a statistically significant difference in either direction (p_{Bonf} 0.354–1.0), and the sign of the effect is inconsistent across backbones (positive in 3/5, negative in 2/5). The effect sizes themselves are small (0.26–0.89) and centered near zero, suggesting a genuine absence of effect rather than an underpowered trend: feature-level distillation does not yield a reliable improvement over span_tiny's main recipe.

5.7.2 Saliency-weighted identity-critical loss

This mechanism reweights the pixel-reconstruction loss by a saliency map derived from identity-relevant image regions, intended to concentrate SR fidelity where it matters most for recognition. Table 12 compares the best-performing saliency weight identified during screening against the main recipe.

Table 12: Saliency-weighted loss ablation vs. span_tiny's main recipe, downstream accuracy, EarVN1.0, $n = 5$ seeds.

Backbone	span_tiny	span_tiny+saliency	Cohen's d	p_{Bonf}
EfficientNet-B0	0.5485	0.5559	0.90	0.3462
GhostNet-100	0.5115	0.5150	0.42	1.0
MobileNetV2	0.5026	0.5053	0.29	1.0
MobileNetV3-Small	0.4423	0.4441	0.11	1.0
ResNet-18	0.4825	0.4800	−0.29	1.0

Again no backbone reaches significance, with small effect sizes (0.11–0.90) and a positive trend in 4/5 backbones that never clears Bonferroni correction. As with KD v2, the saliency-weighted loss does not produce a statistically detectable improvement.

5.7.3 Learned/differentiable block pruning

Rather than span_tiny's fixed choice to keep the first three of six SPABs, this mechanism learns which blocks to keep via a differentiable gating mechanism, with a sparsity penalty pushing the model toward a target size. At the sparsity level that produces an exact match to span_tiny's parameter count (0.230M, 3 blocks retained), Table 13 compares the two block-selection strategies at equal size.

Table 13: Learned block pruning vs. span_tiny's fixed block selection, downstream accuracy at an identical parameter count (0.230M), EarVN1.0, $n = 5$ seeds.

Backbone	span_tiny (fixed)	Learned pruning	Cohen's d	p_{Bonf}
EfficientNet-B0	0.5485	0.5222	−3.71	**0.0035**
GhostNet-100	0.5115	0.4806	−3.54	**0.0041**
MobileNetV2	0.5026	0.4685	−3.00	**0.0078**
MobileNetV3-Small	0.4423	0.4095	−3.34	**0.0052**
ResNet-18	0.4825	0.4575	−5.98	**0.0005**

Unlike the two mechanisms above, this result is unambiguous: learned pruning is significantly worse than span_tiny's fixed, hand-picked block selection in all 5/5 backbones, with large effect sizes throughout (d −3.0 to −6.0) and every $p_{\text{Bonf}} < 0.01$. At an identical parameter count, automatically learning which blocks to discard performs reliably worse here than simply keeping the first three of six – a negative result for the differentiable-pruning mechanism specifically, not for depth compression in general (which Table 3 shows is close to accuracy-neutral when the retained blocks are chosen by hand).

Summary of Section 5.7. None of the three mechanisms tested improve on span_tiny's original design: two (KD v2, saliency-weighted loss) are statistically null in either direction, and one (learned pruning) is significantly negative in every backbone tested. This is reported as a rigorous negative-result ablation study, using the same paired-test, multi-seed, multi-backbone standard as the paper's positive claims, rather than reframed as partial successes. The paper's positive, validated contribution remains span_tiny's original depth-count reduction (Sections 5.1–5.2).

5.8 Extended ablations toward mechanistic and design generality

The results above establish that depth compression is accuracy-neutral for span_tiny specifically. Three further questions were identified to test whether this finding reflects a general mechanism or an architecture-specific coincidence, and whether the specific design choices made for span_tiny are themselves well-supported. All three have now been run to completion, on EarVN1.0, to the same $n = 5$-seed, five-backbone standard as the main result: the first (attention-parameterization) and third (block-removal position) directly, and the second (recognition-embedding-space distillation) by recognizing, on inspection, that it already has an answer from an existing result under a different name, described below.

5.8.1 Attention-parameterization ablation – completed, hypothesis not supported

Section 6 proposes that SPAN tolerates depth reduction because its attention mechanism carries no learned parameters, so removing blocks removes computation but not learned capacity. If this account is correct, an architecture with *learned* attention/modulation should show a measurably larger accuracy cost from an equivalent proportional depth reduction than span_tiny does. SAFMN [4] and SMFANet [5], both using learned modulation mechanisms, were each halved in block count and evaluated under the identical Track B recipe, $n = 5$ seeds, all five recognition backbones (SR-quality rows in Table 5). Table 14 reports, per backbone, Δ_{arch} (that architecture's own full-depth-minus-half-depth accuracy) against Δ_{SPAN} (span_large minus span_tiny, from Table 3), and a paired test on their difference.

Table 14: Attention-parameterization ablation, EarVN1.0, $n = 5$ seeds. $\Delta_{\text{arch}} - \Delta_{\text{SPAN}}$ is the diff-of-diffs (positive = the learned-attention architecture loses *more* from depth-halving than SPAN, supporting the parameter-free account); p_{Bonf} corrected across 5 backbones per architecture.

Backbone	Δ_{SAFMN}	Δ_{SMFANet}	diff-of-diffs (d, p_{Bonf}) SAFMN	diff-of-diffs (d, p_{Bonf}) SMFANet
EfficientNet-B0	0.0066	0.0023	-0.0017, d=-0.14, p=1.0	-0.0060, d=-0.79, p=0.766
GhostNet-100	-0.0021	0.0012	-0.0046, d=-0.54, p=1.0	-0.0013, d=-0.18, p=1.0
MobileNetV2	0.0019	0.0002	0.0000, d=0.00, p=1.0	-0.0016, d=-0.14, p=1.0
MobileNetV3-Small	-0.0009	0.0074	-0.0059, d=-0.33, p=1.0	0.0025, d=0.30, p=1.0
ResNet-18	0.0038	-0.0008	0.0000, d=0.00, p=1.0	-0.0046, d=-0.33, p=1.0

Δ_{SPAN} itself (0.0018–0.0083 across backbones, Table 3) is small throughout, and neither SAFMN nor SMFANet loses significantly more than SPAN does from an equivalent proportional depth reduction in *any* of the 10 backbone×architecture combinations tested (p_{Bonf} 0.766–1.0). If anything, the diff-of-diffs is negative (SPAN loses as much or more) in 7/10 combinations: SAFMN and SMFANet, despite using learned modulation mechanisms, tolerate an equivalent depth reduction just as well as span_tiny does. The parameter-free-attention account proposed in Section 6 therefore finds no support here and is revised there accordingly: depth tolerance at this compression level and resolution appears to be a broader property of lightweight SR architectures under this training recipe, not a consequence specific to SPAN's parameter-free design.

5.8.2 Block-removal position ablation – completed, no significant effect

span_tiny retains the first three of six SPABs (Figure 1), a choice made for implementation convenience rather than validated against alternatives. Two further 3-block variants, holding total block count fixed, test whether removal position matters independently of removal count: one retaining the last three blocks instead (removing blocks 1–3), and one retaining an interleaved subset (blocks 1, 3, 5).

Testing this meaningfully requires more than relabelling a freshly-initialized 3-block network: since span_tiny is trained end-to-end from a random initialization under output-level distillation only, a 3-block student trained this way has no internal notion of "which blocks of a virtual 6-block network it occupies" – three variants trained identically would be expected to differ only by seed noise, not by the position nominally assigned to them. To give position a training signal, each variant instead uses a position-targeted feature-hint loss: the student's final-block output is matched, via a learned 1×1 adapter and the same instance-normalized L_1 formulation already used for the feature-KD term (Section 5.7), against the teacher's output at a specific block index representing the depth that variant is meant to occupy (block 3 for the current first-three-blocks recipe, block 6 for a last-three-blocks recipe, block 5 for an interleaved 1-3-5 recipe). All three variants use this same mechanism, differing only in which teacher block is targeted, so the comparison isolates position specifically rather than confounding it with the presence or absence of positional supervision. The teacher for this ablation is span_large rather than span_baseline/span_official: both are 6-block, but only span_large shares the same re-implementation class as span_tiny (Section 3.2), giving well-defined intermediate block outputs to target; span_official's internal structure is external, third-party code without a guaranteed matching interface.

Table 15 reports downstream identity accuracy for the three variants across all five recognition backbones at $n = 5$ seeds.

Table 15: Block-removal position ablation, downstream accuracy, EarVN1.0, $n = 5$ seeds.

Backbone	keep-first (current)	keep-last	interleaved
EfficientNet-B0	0.5389	0.5417	0.5403
GhostNet-100	0.5027	0.5084	0.5034
MobileNetV2	0.4837	0.4890	0.4820
MobileNetV3-Small	0.4246	0.4240	0.4330
ResNet-18	0.4763	0.4791	0.4740

None of the three pairwise comparisons reaches significance in any backbone (p_{Bonf} 0.353–1.0 across all 15 backbone×pair combinations), with effect sizes small throughout ($|d|$ 0.13–0.89). Keep-last ranks highest in 4/5 backbones, a mild directional trend that never clears Bonferroni correction; MobileNetV3-Small is the exception, favoring the interleaved variant instead. Block-removal position does not detectably affect downstream accuracy at this compression level.

An initial $n = 3$-seed screening pass, run before extending to the full five seeds, had flagged one nominally significant difference (EfficientNet-B0, keep-first vs. interleaved: $d = 8.80$, $p_{\mathrm{Bonf}} = 0.013$). It did not replicate once seeds 44 and 999 were added ($d = -0.13$, $p_{\mathrm{Bonf}} = 1.0$): the added seeds shifted keep-first's mean downward and widened its variance enough to erase the effect entirely. This is the extension protocol working as intended – the same $n = 3 \to n = 5$ seed-extension rule applied throughout this study (Section 5.7) exists specifically to catch small-sample false positives of this kind before they are reported as findings.

5.8.3 Recognition-embedding-space distillation

Ataer-Cansizoglu et al. [6] and the iris-recognition literature reviewed in Section 2 (Nguyen et al. [7], Ribeiro et al. [8]) suggest that SR objectives aligned with a downstream recognition embedding, rather than with pixel or teacher-feature reconstruction, are more likely to transfer to recognition accuracy. This account is directly testable with a loss term matching embeddings extracted from a frozen, pretrained recognition backbone, computed between span_tiny's SR output and the true HR image:

$$\mathcal{L}_{\text{embed}} = 1 - \cos\big(f(\text{SR}(x)),\ f(\text{HR})\big), \tag{1}$$

where $f(\cdot)$ denotes the frozen backbone's penultimate-layer embedding. This is, on inspection, mathematically identical to the identity-loss term already evaluated in Section 5.3 and Section 3.3: both are a cosine loss between a frozen recognition backbone's embeddings of the SR output and the true HR image, using the same backbone (MobileNetV2) as the embedding source. The two configurations this account calls for – pixel plus $\mathcal{L}_{\text{embed}}$ alone, and pixel plus distillation plus $\mathcal{L}_{\text{embed}}$ – are exactly the pixel+identity and pixel+distill+identity ("full") rows already reported in Table 4, not a new experiment. Both fall in the lower-performing cluster there, and the identity-loss term is independently confirmed to significantly reduce accuracy by the $\lambda_{\text{identity}}$ sweep in Section 3.3 ($d = -10.09$). The recognition-aware-objective account therefore does not hold in this setting: unlike the face- and iris-recognition results that motivate it, aligning the SR loss with a frozen recognition backbone's embedding space does not improve, and in fact measurably harms, downstream identity accuracy for ear SR at this compression level.

6 Discussion

Interpreting the trade-off. The design goal of this work was not maximal downstream accuracy but a specific, pre-defined criterion: preserve the ordering `lr` < `span_tiny` < `span_baseline` while substantially reducing parameter count and latency, and in particular keep the compression-specific cost (`span_tiny` vs. `span_baseline`) small and, ideally, statistically undetectable. On EarVN1.0 this cost is non-significant in 4/5 backbones (Table 2) and, on the depth-isolated comparison, in 4/5 backbones as well (Table 3). On AWEx it is non-significant in 5/5 backbones under all three protocols tested – from-scratch, full-fine-tune transfer, and frozen-backbone transfer (Section 5.6) – the single most consistent result in the paper. This forms the paper's central empirical claim.

Why depth reduction is close to accuracy-neutral – revised after testing it directly. An earlier draft of this account proposed that SPAN tolerates depth reduction because its attention mechanism carries no learned parameters: removing SPABs removes attention *computation* but not learned attention *capacity*, since there is none to lose. This account makes a testable directional prediction – an architecture with *learned* attention/modulation should lose more from an equivalent depth cut – and Section 5.8 tested it directly against SAFMN and SMFANet. The prediction was not confirmed: neither architecture loses significantly more than SPAN does from a proportional depth halving in any of 10 backbone×architecture combinations, and the diff-of-diffs is negative (SPAN losing as much or more) in 7/10. The parameter-free-attention account is therefore not supported by this test and is retracted as the explanation. A more likely account, consistent with both this ablation and the loss-ablation result below, is that at this extreme compression level and input resolution (20×20 px), the recognition-relevant signal in an ear crop is concentrated in relatively coarse, low-frequency structure that a shallower network already captures adequately, regardless of its attention mechanism's parameterization – making depth reduction broadly tolerable across this class of lightweight SR architectures rather than a property specific to SPAN. This is offered as a plausible revised account, not a proven mechanism; distinguishing it from other candidate explanations is left to future work (Section 8).

Why the distillation-term ablation is negative while the identity-loss ablation is positive, and why neither undermines the main claim. Section 5.3 finds two distinct results that should not be conflated: the identity-loss term *significantly reduces* downstream accuracy (3/6 pairwise comparisons significant, 1/6 at trend level, corroborated independently by the $\lambda_{\text{identity}}$ sweep in Section 3.3, d=−10.09), while the distillation term's own marginal contribution over pixel-only training remains statistically indistinguishable (d=−0.40, p_{Bonf}=1.0). For the latter, null result, three explanations seem plausible: the pixel-reconstruction signal alone may already be close to sufficient at this parameter scale; seed-to-seed variance may exceed the true effect size; or, as Zhang et al. [16] argue for SR distillation generally, a teacher's own output is only an approximation of the true HR image, which caps how much signal the distillation term can supply. The data collected here cannot distinguish between these. Neither result undermines the paper's central claim, which concerns architectural depth, not loss-function choice; span_tiny's training recipe (Track B, pixel+distillation, $\lambda_{\text{identity}}$=0) remains justified independently by its role in stabilizing convergence for other architectures in the contextual comparison (Section 5.4) and by the identity-loss ablation's own negative finding for that specific term.

SMFANet convergence case study. SMFANet failed to converge well under pixel-loss-only supervision (20.906 dB, Table 5) but recovered strongly under distillation supervision (+5.4 dB). The marginal contribution of distillation to span_tiny's own downstream accuracy was not statistically detectable, but this case shows that distillation supervision is not uniformly redundant across architectures at this extreme input resolution. Some architectures appear to depend on it to converge at all.

Position relative to other lightweight SR architectures – stronger than raw image quality suggests. Beyond the SPAN family, span_tiny is second-fastest and third-smallest among six compared architectures and ranks third-lowest on raw PSNR/SSIM (Table 5) – but on downstream recognition accuracy, the metric this paper is organized around, span_tiny is never significantly beaten by any of the other four architectures across any of the five recognition backbones, is statistically tied with the two most accurate ones (RLFN-adapted, SAFMN), and significantly *outperforms* the two least accurate ones (ECBSR: 4/5 backbones significant, 1/5 trend; SMFANet: 2/5 significant, 1/5 trend) (Table 6). This is a materially stronger position than the raw image-quality ranking alone would suggest, and it is worth stating plainly what it does and does not mean: it is not evidence that span_tiny is the best lightweight SR architecture in absolute terms (SAFMN and RLFN-adapted remain fully competitive, at higher latency for SAFMN and a comparable one for RLFN-adapted), but it is evidence that raw PSNR/SSIM ranking, on its own, would have misidentified span_tiny as accuracy-inferior to ECBSR and SMFANet when the downstream task says otherwise. The re-implementation's timing broadly matches the original SPAN paper's comparison against SAFMN on standard SR benchmarks (≈13 ms/28.66 dB vs. ≈72 ms/≈28.60 dB on Set14 ×4 in [1]), an independent data point supporting the general efficiency trade-off observed here, alongside the new evidence that this trade-off does not come at a downstream-accuracy cost relative to the weaker-PSNR alternatives.

Diverging statistical power between datasets, and between transfer protocols. The weaker and more protocol-dependent statistical significance of `SR` > `lr` on AWEx (Section 5.6) compared to EarVN1.0 (Table 2, significant in 5/5 backbones for both span_tiny and span_baseline) follows from AWEx's smaller per-subject training sample (≈6.7 images/subject, versus substantially more on EarVN1.0), not from a weaker or contradictory effect. It is not evidence of no effect. This applies specifically to the *secondary* question (does SR help over no-SR); the paper's *central* claim, that compression itself is safe, is the more consistent finding across both datasets and is confirmed under all three protocols tested on AWEx with 0/5 backbones showing a significant cost in any of them – if anything, this claim is *more* thoroughly replicated on AWEx (three independent protocols) than the secondary question is.

Unresolved and resolved observations. Two anomalies flagged at an earlier stage of this study are resolved under the corrected experimental setup and are not carried forward: an off-trend ResNet-18 result on the AWEx from-scratch track (Section 5.6), and span_large's anomalously low PSNR on AWEx (previously ≈19 dB; now 24.4 dB, in line with the other three core architectures, Table 8). One new finding is reported as an investigated boundary condition rather than an unexplained anomaly: the reversed ordering on the EarVN1.0 real-low-resolution holdout (Section 5.1), traced to every holdout image's native resolution falling below the SR models' trained input size. One genuine open item remains: ResNet-18's inconsistent secondary-question evidence across all three AWEx protocols (Section 5.6.2), for which no confirmed explanation exists and which is carried forward as a limitation rather than a subject for speculation.

7 Limitations

Several limitations concern AWEx specifically, and none of them affects the paper's central compression-safety claim, which is confirmed there with zero of five backbones showing a significant cost under all three protocols tested. AWEx's smaller per-subject sample size ($\approx$6.7 training images/subject) does limit statistical power for the secondary question of whether SR helps over no-SR at all (Section 5.6): effects are directionally positive in most backbone$\times$protocol combinations but do not always clear the significance threshold, and which backbones reach significance varies by protocol; this should be read as underpowered rather than null (Section 6). ResNet-18 is the clearest case of this: across all three AWEx protocols it shows the weakest and least consistent evidence that SR helps at all, including one instance (frozen-backbone transfer) where span_tiny falls nominally, non-significantly, below no-SR, and no confirmed explanation exists for why this backbone behaves differently from the other four on this dataset. The AWEx depth ablation (span_tiny vs. span_large) also runs at a reduced sample size, $n = 3$ seeds rather than the $n = 5$ used elsewhere, so its non-significant result in all five backbones carries correspondingly less power, even though it agrees with the EarVN1.0 finding. Separately, the six-architecture context comparison, the attention-parameterization ablation, and the three additional-mechanism ablations were all validated on EarVN1.0 only in this phase of the study; whether span_tiny's competitive standing against other architectures, the null result for the parameter-free-attention account, and the negative or null findings for KD v2, the saliency-weighted loss, and learned pruning replicate on AWEx is untested and left as a direction for future work rather than assumed. Finally, AWEx is evaluated with a custom closed-set, per-subject train/validation/test split rather than its official open-set split, because this study addresses closed-set identity classification rather than the open-set verification problem that official split was designed for; this choice is considered appropriate for the task at hand and is stated explicitly to avoid any impression of an undisclosed deviation from a standard protocol.

Two further ablations on EarVN1.0 are validated at reduced scope. The loss-component ablation (Section 5.3) covers a single backbone, MobileNetV2; whether the identity-loss finding and the distillation null result hold for the other four backbones, or on AWEx, remains untested, and the independent $\lambda_{\text{identity}}$ sweep (Section 3.3) corroborates the identity-loss direction on the same backbone without extending that coverage either. The unmodified `rlfn` baseline (distinct from `rlfn_adapted`, the primary RLFN reference used throughout) has $n = 3$ seeds at the time of writing, short of the $n = 5$ used elsewhere.

Two measurement-level limitations apply throughout. Latency figures (Table 5) come from the development machine rather than a representative resource-constrained device such as a Raspberry Pi or a phone SoC, so they should be read as relative comparisons under controlled conditions rather than deployment-ready benchmarks. PSNR/SSIM/LPIPS (Table 5, Table 8) come from a single training run per configuration ($n = 1$), with no confidence interval, unlike downstream recognition accuracy, which is multi-seed throughout.

Finally, the paper's main claim is validated specifically for the synthetic bicubic-downsampling regime defined in Section 3, and this scope should be stated plainly rather than assumed to extend further: an auxiliary check on genuinely low-resolution real-world images (Section 5.1) shows the opposite ordering, traced to a resolution mismatch between these images (all natively below the SR models' trained input size) and the training-time degradation. This is a boundary condition on where the claim has been shown to hold, not a refutation of it, but it means this paper does not provide evidence that span_tiny or span_baseline improves recognition on arbitrary real-world low-resolution ear images outside the synthetic-degradation regime studied here – that check itself stands at $n = 1$ (173 images, one backbone), with no multi-seed confirmation.

8 Conclusion

This paper asked a narrower question than most efficient-SR work: not whether a compressed model is the most accurate option available, but whether compressing SPAN's depth by half still leaves a model worth using ahead of no SR at all, without paying a real accuracy price for the compression itself. Across five recognition backbones and two independent datasets, span_tiny answers both parts of that question affirmatively, and more strongly than a coarser reading of the numbers alone would suggest. `span_tiny` beats `no-SR` with statistical significance in 5/5 backbones on EarVN1.0; the specific cost of compression (`span_tiny` vs. `span_baseline`) is non-significant in 4/5 backbones there and in 5/5 backbones under all three evaluation protocols tested on AWEx (from-scratch, full-fine-tune transfer, frozen-backbone transfer) – the most consistently replicated finding in the paper. The depth ablation isolating block count alone corroborates this directly: span_tiny and its uncompressed-depth counterpart, span_large, are statistically indistinguishable in 4/5 backbones on EarVN1.0 and 5/5 on AWEx. Against five other lightweight SR architectures, span_tiny is never significantly beaten on downstream accuracy and significantly outperforms the two weakest of them, despite ranking low on raw PSNR/SSIM – a concrete demonstration that pixel-fidelity metrics are an imperfect stand-in for the metric this paper actually cares about.

The evidence is reported with equal clarity where it runs out or points against an earlier hypothesis. Three further mechanisms proposed to improve on span_tiny – a richer distillation loss, a saliency-weighted loss, and learned block pruning – were each validated to the same statistical standard as the main result and are reported as negative or null, with learned pruning significantly worse than span_tiny's fixed block selection in every backbone tested. An ablation designed to test this paper's own proposed explanation for why depth reduction is tolerable – SPAN's parameter-free attention – returned a result against that explanation, and the account is revised rather than defended. An auxiliary check on genuinely real, low-resolution images reversed the main claim's ordering, and rather than being left unexplained, this was traced to a specific, verifiable cause (a resolution mismatch between these images and the synthetic degradation used in training) and is reported as a boundary condition on the paper's scope. One anomaly remains open: ResNet-18's inconsistent secondary-question evidence on AWEx, across all three protocols, lacks a confirmed explanation. None of this changes the central result, but a reader should be able to see exactly how far the evidence goes, where a hypothesis was tested and did not survive, and where it stops.

Extending this work would involve latency measured on an actual edge device rather than development hardware, confidence intervals for the SR-quality metrics (currently single-run), extending the six-architecture context comparison and the three additional-mechanism ablations to AWEx to test whether they replicate cross-dataset, and, if a suitable third dataset becomes available, an additional independent check on generalization beyond EarVN1.0 and AWEx. Two further questions raised in Section 5.8 about span_tiny's specific design choices have since been resolved rather than left open: a block-removal position ablation, testing whether the specific choice of retained blocks was well-supported, found no significant effect of removal position at $n = 5$ seeds across all five backbones; and a candidate recognition-embedding-space distillation loss, motivated by the recognition-aware SR literature reviewed in Section 2, turned out on inspection to be mathematically identical to the identity-loss term already evaluated in Section 5.3, an account that does not hold in this setting.

Declarations

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Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

Data availability EarVN1.0 is available via Mendeley Data (https://doi.org/10.17632/yws3v3mwx3.4) under the terms specified by the original authors [9], which reserve all rights to the database and prohibit commercial use or redistribution; access is therefore via the original repository only, not bundled with this paper. AWE/AWEx access requires a signed request form submitted to the University of Ljubljana ear-recognition research group, for research use only.

Code availability The code that supports the findings of this study is available from the corresponding author upon reasonable request.

Ethics approval This study did not involve the collection of any new data from human subjects. All experiments were conducted on publicly available, previously published datasets (EarVN1.0, AWE/AWEx) that were de-identified by their original authors under their own data-collection protocols; no additional institutional ethics approval was required for this secondary use.

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